

## Homework 4: Static Methods

1.

```
private static boolean pointIsInCircle(double xCoord, double yCoord) {
    double centerX = 1.0;
    double centerY = 1.0;
    double radius = 1.0;
    double distance = Math.pow(xCoord - centerX, 2) + Math.pow(yCoord - centerY, 2);
    return distance <= Math.pow(radius, 2);
}
```

2.

```
private static int numberOfPointsInCircle(int n) {
    int count = 0;
    for (int i = 0; i < n; i++) {
        double xCoord = Math.random() * 2.0;
        double yCoord = Math.random() * 2.0;
        if (pointIsInCircle(xCoord, yCoord)) {
            count = count + 1;
        }
    }
    return count;
}
```

3.

```
public static void main(String[] args) {
    SimpleReader input = new SimpleReader1L();
    SimpleWriter output = new SimpleWriter1L();
    output.print("Number of points: ");
    int pointNumber = input.nextInteger();
    int pointInCircle = numberOfPointsInCircle(pointNumber);
    double estimateArea = 4.0 * pointInCircle / pointNumber;
    output.println("Estimate of area: " + estimateArea);
    input.close();
    output.close();
}
```